

Class 26 Math Induction

Goal

Learn to use the technique of mathematical induction.

Induction proofs are typically used to prove statements that depend on integers.

Examples

Weak Principle of Math Induction

Example 1 Prove $\sum_{i=1}^n 2i = n^2 + n \quad \forall n \geq 1$

Base Case :

Weak induction step

[show if $p(k)$ is true, then $p(k+1)$ is true]

Conclusion: Since the statement is true for $n=1$ and is true for $n=k+1$ whenever true for $n=k \geq 1$ we know the statement is true for all integers $n \geq 1$

Example 2 Prove $2^n < (n+2)! \quad \forall n \geq 0$

Base Case :

Conclusion: Since the statement is true for $n=0$ and is true for $n=k+1$ whenever true for $n=k \geq 0$, we know the statement is true for all integers $n \geq 0$.